AMR Motion-Planning Papers

PDF collection and clickable reading index | 1 November 2026

Scope: onboard perception-only robot autonomy is the default. Peer trajectories, server knowledge, and reservations are optional future research. PDFs linked here do not imply that all papers have been downloaded into the archive.

What is physically included: five previously provided full papers. **Other references:** online author/publisher open copies linked below. On your own Ubuntu computer run `python3 download_open_papers.py --all` to fetch the publicly accessible PDFs.

Reading order

1. OA-MPC - Blind corners, conservative reachable occupancy, fallback stopping

2. ORCA-DD / acceleration-aware ORCA - Independent, noncommunicating differential-drive robot avoidance

3. Werling + TEB 2012 - Frenet lateral offsets, longitudinal timing, model constraints

4. Schöneberg + Trepella + SICNav - Time-indexed unknown-agent predictions and interaction-conditioned MPC

5. D4orm / benchmarking papers - Candidate refinement, baseline comparisons, scenario tools

Already included as PDFs

01_Werling_2010_Frenet.pdf - Frenet trajectory generation

02_Roesmann_2012_TEB.pdf - Original timed elastic band

03_Roesmann_2017_TEB.pdf - Kinodynamic trajectory refinement (car-like)

04_Chung_2022_DTEB.pdf - Distributed timed trajectories (optional cooperation)

05_VanDenBerg_ORCA.pdf - Independent reciprocal collision avoidance

Additional direct PDF links

Sensor-only core

P0 Occlusion-Aware MPC for Guaranteed Safe Robot Navigation With Unseen Dynamic Obstacles - Unseen agents, reachable occupancy and terminal stop feasibility

P0 Smooth and Collision-Free Navigation for Multiple Robots Under Differential-Drive Constraints - Reciprocal motion without communication, differential-drive kinematics

P0 Smooth Coordination and Navigation for Multiple Differential-Drive Robots - Acceleration-aware reciprocal collision avoidance

P1 SIPP: Safe Interval Path Planning for Dynamic Environments - Space-time route availability with dynamic obstacles

P0 Trajectory Planning with MPC for Obstacle Avoidance Considering Prediction Uncertainty - Uncertainty-aware dynamic obstacle prediction with ROS 2/Nav2

Interaction prediction

P0 Navigating the Crowd: Non-linear MPC with Social Forces Dynamics for Human-Aware Robot Navigation - Robot-conditioned pedestrian motion inside MPC; Nav2 plugin

P0 SICNav: Safe and Interactive Crowd Navigation using MPC and Bilevel Optimization - Robot-conditioned prediction via ORCA based human-response model

P1 SICNav-Diffusion: Safe and Interactive Crowd Navigation With Diffusion Trajectory Predictions - Multimodal human trajectories and coupled planning

P1 Trajectron++: Dynamically-Feasible Trajectory Forecasting With Heterogeneous Data - Map-aware multi-hypothesis prediction; optional ego conditioning

P2 Social LSTM: Human Trajectory Prediction in Crowded Spaces - Foundational model of human-agent interaction

Trajectory planning

P1 D4orm: Multi-Robot Trajectories with Dynamics-aware Diffusion Denoised Deformations - Anytime dynamical trajectory refinement including differential-drive models

P1 Reciprocal n-body Collision Avoidance - Foundational reciprocal velocity-obstacle analysis

Optional group coordination

P2 Priority Inheritance with Backtracking for Iterative Multi-agent Path Finding - Local priority concepts; assumptions do not fit arbitrary factory agents

P2 Distributed Timed Elastic Band Planner: Trajectory Sharing and Collision Prediction - Communicated timed plans, future optional enhancement

P2 Robust MADER: Decentralized Multiagent Trajectory Planner Robust to Communication Delay - Delayed intent/commitment; optional robots with communications

P2 LF: Online Multi-Robot Path Planning Meets Optimal Trajectory Control - Global-knowledge multi-robot architecture comparison

P2 Multi-Agent Pathfinding with Continuous Time - Continuous time conflict search; only if future coordination included

P2 Real-Time Multi-Robot Motion Planning with Safe-Interval Search and Learning-Guided Repair - Learning-guided congestion repair; requires coordinated robots

P2 Congestion Mitigation Path Planning for Large-Scale Multi-Agent Navigation - Coarse traffic-aware routing, requires multi-agent context

P2 EGO-Swarm: A Fully Autonomous and Decentralized Quadrotor Swarm System - Local topology alternatives, aerial-model and communication caveats

Benchmarking

P1 HuNavSim: A ROS 2 Human Navigation Simulator for Benchmarking Human-Aware Robot Navigation - Reusable difficult pedestrian behavior scenarios for ROS 2

P1 Arena 4.0: A ROS2 Development and Benchmarking Platform - Human-aware scenario generation, large PDF

P1 Arena 5.0: A ROS2 Social Navigation Benchmark - Modern scenario and evaluation tools, large PDF

Extended literature catalog (69 references)

The links below are research pages or full-text pages as listed in the existing survey; they are not all direct PDFs. For papers without a confirmed publicly available full PDF, use the publisher or author page. The catalog includes optional centralized/cooperative literature for comparison only.

Trajectory generation / control

Werling et al. — Optimal Trajectory Generation for Dynamic Street Scenarios in a Frenet Frame (2010; P0)

Rösmann et al. — Trajectory Modification Considering Dynamic Constraints of Autonomous Robots (2012; P1)

Rösmann et al. — Kinodynamic Trajectory Optimization and Control for Car-Like Robots (2017; P2)

Xin et al. — Distributed Model Predictive Contouring Control for Real-Time Multi-Robot Motion Planning (2022; P0)

Schöneberg et al. — Trajectory Planning with MPC for Obstacle Avoidance Considering Prediction Uncertainty (2025; P1)

Sun et al. — Integrated Multi-AMR CL-CBS and MPC-APF in Warehousing (2025; P2)

Zhang et al. — D4orm: Multi-Robot Trajectories with Dynamics-aware Diffusion Denoised Deformations (2025; P1)

Shankar, Okumura and Prorok — LF: Online Multi-Robot Path Planning Meets Optimal Trajectory Control (2025; P0)

Samavi et al. — SICNav-Diffusion: Safe and Interactive Crowd Navigation With Diffusion Trajectory Predictions (2025; P1)

Trepella et al. — Navigating the Crowd: Non-linear MPC with Social Forces Dynamics (2026; P0)

Sun et al. — A Decoupled Time-Optimized Multi-Robot Path Parameterization Method (2026; P1)

Qu et al. — Safety Corridor and Pedestrian Prediction Integrated Sequential MPC (2026; P1)

Distributed multi-robot avoidance

van den Berg et al. — Reciprocal n-body Collision Avoidance (ORCA) (2011; P1)

Snape et al. — ORCA for Differential-Drive Robots (ORCA-DD) (2010; P1)

Zhou et al. — Fast On-line Collision Avoidance for Dynamic Vehicles Using Buffered Voronoi Cells (2017; P2)

Wang and Schwager — Distributed Collision Avoidance with Probabilistic Buffered Voronoi Cells (2019; P2)

Zhu, Brito and Alonso-Mora — Decentralized Probabilistic Multi-Robot Collision Avoidance with Buffered Uncertainty-aware Voronoi Cells (2022; P1)

Zhou et al. — EGO-Swarm: Fully Autonomous and Decentralized Quadrotor Swarm (2021; P1)

Tordesillas et al. — MADER: Trajectory Planner in Multiagent and Dynamic Environments (2022; P1)

Kondo et al. — Robust MADER: Decentralized Multiagent Trajectory Planner Robust to Communication Delay (2024; P0)

Chung et al. — Distributed Timed Elastic Band (DTEB) Planner (2022; P0)

IFAC — Distributed MPC for Safety-Critical Obstacle Avoidance with Social Preferences (2025; P2)

Muhammed et al. — Real-time Decentralized MPC for Cooperative Multi-Robot Object Transport (2026; P2)

Traffic scheduling / MAPF

Phillips and Likhachev — SIPP: Safe Interval Path Planning for Dynamic Environments (2011; P0)

Ali and Yakovlev — Safe Interval Path Planning with Kinodynamic Constraints (SIPP-IP) (2023; P0)

Kasaura et al. — Prioritized SIPP with Continuous Time on 2D Roadmaps (PSIPP/CTC) (2022; P0)

Andreychuk et al. — Multi-Agent Pathfinding with Continuous Time (CCBS) (2019; P1)

Tan et al. — Robust Multi-Agent Pathfinding with Continuous Time (2024; P0)

Okumura et al. — Priority Inheritance with Backtracking (PIBT) (2019; P1)

Li et al. — Lifelong MAPF in Large-Scale Warehouses (RHCR) (2021; P1)

Li et al. — MAPF-LNS2: Fast Repairing via Large Neighborhood Search (2022; P1)

Huang et al. — Anytime MAPF via ML-Guided Large Neighborhood Search (2022; P3)

Okumura — LaCAM: Search-Based Algorithm for Quick Multi-Agent Pathfinding (2023; P2)

Okumura — Improving LaCAM for Scalable Eventually Optimal MAPF (2023; P2)

Kato et al. — Congestion Mitigation Path Planning for Large-Scale Multi-Agent Navigation (2025; P1)

Amir et al. — ReCoDe: RL-Based Dynamic Constraint Design for Multi-Agent Coordination (2025; P2)

Okumura et al. — Concrete Multi-Agent Path Planning Enabling Kinodynamically Aggressive Maneuvers (2026; P0)

Kim et al. — A Scalable Framework for Lifelong MAPF with Asynchronous Actions (2026; P0)

Zheng et al. — Learning-guided Prioritized Planning for Lifelong MAPF in Warehouse Automation (2026; P2)

Klan■ar et al. — Elevating Priorities for Efficient Lifelong Multi-AGV Pathfinding on Roadmaps (2026; P1)

Tang et al. — Greedy Priority Inheritance with Backtracking (2026; P2)

Arita and Okumura — Local Guidance for Configuration-Based Multi-Agent Pathfinding (2026; P2)

Jain et al. — Graph Attention-Guided Search for Dense Multi-Agent Pathfinding (LaGAT) (2026; P3)

Arita and Okumura — Lifelong LaCAM with Local Guidance (2026; P2)

Kumar et al. — Real-Time Multi-Robot Motion Planning with Safe-Interval Search and Learning-Guided Repair (2026; P1)

Huang et al. — SMART: Scalable Multi-Agent Reasoning and Trajectory Planning in Dense Environments (2026; P2)

Guo et al. — Collision Risk Index and Dynamic Priority in Multi-Robot Cooperative Planning (2026; P3)

Prediction / interaction

Salzmann et al. — Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data (2020; P2)

Sun et al. — M2I: From Factored Marginal to Interactive Trajectory Prediction (2022; P2)

Alahi et al. — Social LSTM: Human Trajectory Prediction in Crowded Spaces (2016; P3)

Gupta et al. — Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks (2018; P3)

Yang et al. — IA-LSTM: Interaction-Aware LSTM for Pedestrian Trajectory Prediction (2024; P3)

Samavi et al. — SICNav: Interactive Crowd Navigation with Bilevel Optimization (2024; P0)

Autoware — Map-Based Prediction Implementation (versioned documentation) (2024; P0)

Qiu and Liu — Intent-Aware Spatiotemporal Risk Field for Multi-AGV Planning (2026; P1)

Meng et al. — Risk-Aware Multi-Modal Pedestrian Prediction with Scene-Attentive Social Attention (2026; P3)

Pan et al. — Mixed Crowd Navigation: Perception, Interaction, Planning, and Control (survey) (2026; P2)

Safety / occlusion

Firoozi et al. — OA-MPC: Occlusion-Aware MPC for Guaranteed Safe Robot Navigation with Unseen Dynamic Obstacles (2025; P0)

Haraldsen et al. — Theoretical Analysis of Velocity Obstacle Method for Nonholonomic Vehicles (2024; P1)

Zheng et al. — Occlusion-Aware Contingency Safety-Critical Planning for Autonomous Driving (2026; P1)

Ye et al. — Risk-Aware and Socially Compliant Topological Motion Planning Under Uncertainty (2026; P2)

Safety-Critical Coordination of Legged Robots via Forward Reachable Set CBFs (2024; P3)

Social Robot Navigation through Constrained Optimization: Uncertainty-Based Objectives and Constraints (2024; P2)

Benchmark / implementation

Shcherbyna et al. — Arena 4.0 ROS2 Navigation Benchmark Platform (2024; P1)

Shcherbyna et al. — Arena 5.0 ROS2 Social Navigation Benchmark (2026; P1)

Pérez-Higueras et al. — HuNavSim ROS 2 Human Navigation Simulator (2023; P1)

Social Robot Navigation: A Review and Benchmarking of Learning-Based Methods (2025; P2)

Bridging Requirements, Planning, and Evaluation: A Review of Social Robot Navigation (2024; P2)

NVIDIA/ROS Nav2 — MPPI Controller Jazzy Documentation (2026; P0)